

Human Review: Uniform Hölder nonexpansive maps on decomposable balls

Original automatic proof: `solution_packet.pdf`

Checked date: 14 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears correct and the method is sound. The key point is that a previous packet in the run had already produced the required uniformly Lipschitzian fixed-point-free map with null minimal displacement on B_X for every decomposable Banach space

$$X = Y \oplus Z,$$

where both summands are infinite-dimensional. Once this input is available, the present argument is a clean scaling step which converts uniformly Lipschitzian dynamics into uniformly α -Hölder nonexpansive dynamics.

There is also a bibliographic typo in the packet: the source paper is listed as being by Cleon S. Barroso and Valdir Ferreira, whereas the paper *Hölder-contractive mappings, nonlinear extension problem and fixed point free results* is authored only by Cleon S. Barroso.

Scaling Argument

Starting with $U : B_X \rightarrow B_X$ such that

$$\text{Fix}(U) = \emptyset, \quad d(U, B_X) = 0, \quad M := \sup_{n \geq 1} \text{Lip}(U^n) < \infty,$$

the proof compresses the dynamics into a small ball rB_X . The scaled map is

$$S(z) = rU(z/r), \quad z \in rB_X.$$

It has the same fixed-point-free and null-minimal-displacement properties, and its iterates remain uniformly Lipschitzian.

The map on the whole unit ball is then

$$V = SR_r,$$

where $R_r : B_X \rightarrow rB_X$ is the radial retraction. Since R_r is 2-Lipschitz and $S(rB_X) \subseteq rB_X$, one obtains

$$V^n = S^n R_r$$

and therefore

$$\|V^n x - V^n y\| \leq \min\{2M\|x - y\|, 2r\}.$$

Choosing r sufficiently small gives

$$\min\{2Md, 2r\} \leq d^\alpha \quad (0 \leq d \leq 2),$$

which is exactly the uniformly α -Hölder nonexpansive estimate

$$\|V^n x - V^n y\| \leq \|x - y\|^\alpha \quad (n \geq 1).$$

Scope of the Result

The automatic packet correctly observes that the result applies to spaces isomorphic to Hilbert space and to spaces with an unconditional basis, because these are decomposable in the required sense.

However, the packet missed one important consequence: the James space case is also covered. The James space is decomposable, and therefore it falls under the same theorem. Thus the argument should be recorded not only as answering the Hilbert-isomorphic and reflexive-unconditional-basis cases, but also as giving the corresponding affirmative answer for James space.

Review Outcome

Archive this packet as an apparently correct solution for all decomposable Banach spaces covered by the input uniformly Lipschitzian construction. In particular, it gives affirmative answers in the Hilbert-isomorphic case, the unconditional-basis case, and the James-space case. The remaining genuinely unresolved part, from the perspective of this packet, is the arbitrary reflexive case without an available decomposable uniformly Lipschitzian construction.